

Discovery of Skill Switching Criteria for Learning Agile Quadruped Locomotion

Wanming Yu
Oxford Robotics Institute
University of Oxford
wanming@robots.ox.ac.uk

Fernando Acero
Department of Computer Science
University College London
fernando.acero@ucl.ac.uk

Vassil Atanassov
Oxford Robotics Institute
University of Oxford
vassilatanassov@robots.ox.ac.uk

Chuanyu Yang
Amigaga Technology Co. Ltd
ycy0char@gmail.com

Ioannis Havoutis
Oxford Robotics Institute
University of Oxford
ioannis@robots.ox.ac.uk

Dimitrios Kanoulas
Department of Computer Science
University College London
d.kanoulas@ucl.ac.uk

Zhibin Li
Department of Computer Science
University College London
alex.li@ucl.ac.uk

Abstract

This paper develops a hierarchical learning and optimization framework that can learn and achieve well-coordinated multi-skill locomotion. The learned multi-skill policy can switch between skills automatically and naturally in tracking arbitrarily positioned goals and recover from failures promptly. The proposed framework is composed of a deep reinforcement learning process and an optimization process. First, the contact pattern is incorporated into the reward terms for learning different types of gaits as separate policies without the need for any other references. Then, a higher level policy is learned to generate weights for individual policies to compose multi-skill locomotion in a goal-tracking task setting. Skills are automatically and naturally switched according to the distance to the goal. The proper distances for skill switching are incorporated in reward calculation for learning the high level policy and updated by an outer optimization loop as learning progresses. We first demonstrated successful multi-skill locomotion in comprehensive tasks on a simulated Unitree A1 quadruped robot. We also deployed the learned policy in the real world showcasing trotting, bounding, galloping, and their natural transitions as the goal position changes. Moreover, the learned policy can react to unexpected failures at any time, perform prompt recovery, and resume locomotion successfully. Compared to discrete switch between single skills which failed to transition to galloping in the real world, our proposed approach achieves all the learned agile skills, with smoother and more continuous skill transitions.

Keywords: Multi-skill locomotion; Deep reinforcement learning; Hierarchical learning and optimization; Gait transitions; Skill switching; Legged locomotion; Robot learning

发现学习敏捷四足运动的技能切换准则

Wanming Yu
牛津大学牛津机器人研究所
wanming@robots.ox.ac.uk

Fernando Acero伦敦大学
学院计算机科学系
fernando.acero@ucl.ac.uk

Vassil Atanassov牛津大学牛津
机器人研究所
vassilatanassov@robots.ox.ac.uk

Chuanyu Yang
Amigaga科技有限公司
ycy0char@gmail.com

Ioannis Havoutis牛津
大学机器人研究所
ioannis@robots.ox.ac.uk

Dimitrios Kanoulas
伦敦大学学院计算机科学系
d.kanoulas@ucl.ac.uk

Zhibin Li
伦敦大学学院计算机科学系
alex.li@ucl.ac.uk

摘要

本文开发了一个分层学习与优化框架，能够学习并实现良好协调的多技能运动。所学习的多技能策略能够在跟踪任意定位目标时自动且自然地切换技能，并能迅速从故障中恢复。所提出的框架由深度强化学习过程和优化过程组成。首先，将接触模式纳入奖励项中，以学习不同类型的步态作为独立策略，无需任何其他参考。然后，学习一个高层策略，为各个策略生成权重，以在目标跟踪任务设置中组合多技能运动。技能根据与目标的距离自动且自然地切换。技能切换的适当距离被纳入奖励计算中，用于学习高层策略，并由外部优化循环随着学习进展进行更新。我们首先在模拟的Unitree A1四足机器人上展示了在综合任务中成功的多技能运动。我们还将学习到的策略部署到现实世界中，展示了随着目标位置变化的小跑、跳跃、疾驰及其自然转换。此外，学习到的策略能够随时对意外故障做出反应，执行快速恢复，并成功恢复运动。与在现实世界中未能成功转换到疾驰的单一技能之间的离散切换相比，我们提出的方法实现了所有学习到的敏捷技能，且技能转换更平滑、更连续。

关键词：多技能运动；深度强化学习；分层学习与优化；步态转换；技能切换；腿式运动；机器人学习

1 Introduction

Animals have evolved highly efficient movement strategies. Mimicking these can improve legged locomotion in terms of agility, stability, and adaptivity (Fig. 1). In particular, animals learn to switch between motor skills swiftly according to tasks and surroundings. For instance, horses switch to different gait patterns as speed changes (Hoyt and Taylor, 1981). However, reproducing multiple gaits and their dynamically feasible transitions on legged robots remains challenging in the robot learning and control community. Besides, the ability to recover from various failures, which is of vital interest for successful and resilient real-world deployment, is not yet well studied in multi-skill locomotion. A multi-skill framework has the ability to recover from failures during locomotion (Yang et al., 2020), however, it does not show more dynamic gaits other than trotting. Although fall recovery has been studied in several previous works (Hwangbo et al., 2019; Castano et al., 2019), it is learned as a single skill but cannot be combined with other skills.

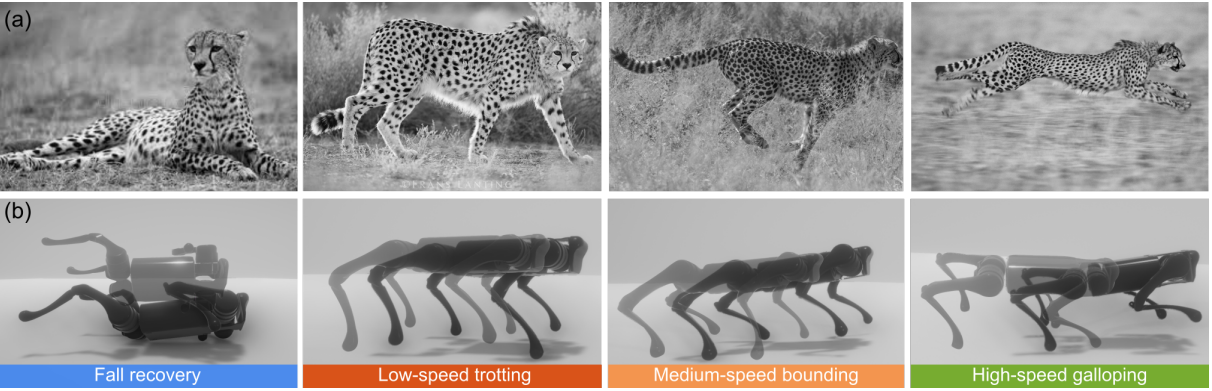


Figure 1: Coordination and gait transitions in quadrupedal animal and robot on demand of increasing speed. (a) Cheetah’s changing gaits at increasing speed. (b) *A1* quadruped robot’s fall recovery, trotting, bounding and galloping skills using our multi-skill policy.

Reproducing gait patterns. Behavior cloning or imitation learning approaches have been applied to reproduce various gaits from reference motions. Reference motions can be captured from animal locomotion which can be limited in variety or generated from model predictive controllers (Reske et al., 2021) which require domain knowledge and computing efficiency. In general, such approaches can be difficult to scale beyond the dataset, lacking robustness and diversity of the learned behaviors. There have been attempts to model or learn parameterized control policy to achieve different styles of walking. A phase-guided controller is used to learn gait transitions between walking, trotting, pacing and bounding on a quadrupedal robot (Shao et al., 2021). A single policy is learned to control various gaits with variable footswing, posture and speed (Margolis and Agrawal, 2023). However, such approaches need to handcraft control-specific behavior parameters, which need domain knowledge and are not intuitive to switch gaits.

Generative models in multi-skill locomotion. Recent advances in generative models have achieved low-level control of locomotion gaits on quadrupedal robots. Variational autoencoders have been applied to learn a disentangled and 2D latent representation across locomotion gaits with respect to footswing heights and lengths (Mitchell et al., 2024). Given desired gait type and swing characteristics, it achieved the low-level control of trotting, crawling and pacing on quadrupedal robots. Besides, diffusion models have shown the capabilities of achieving multi-skill locomotion control by a single policy, such as trotting, hopping, pacing, walking and running (Huang et al., 2024), and walking, crawling and their transitions (O’Mahoney et al., 2024). However, these generative models require expert demonstrations, and the performance is dependent on the quality of the dataset. Moreover, the gait type was conditioned on a certain input to the control policy. In contrast, the gait type was autonomously discovered by our proposed hierarchical framework covering both high-level and low-level multi-skill locomotion control.

Foundation models in legged locomotion. Applying foundation models in robot learning applications

1 引言

动物进化出了高效的运动策略。模仿这些策略可以在敏捷性、稳定性和适应性方面改善腿式运动（图1）。特别是，动物能够根据任务和环境快速切换运动技能。例如，马会随着速度的变化切换到不同的步态模式（Hoyt和Taylor，1981）。然而，在腿式机器人上复现多种步态及其动态可行过渡，在机器人学习与控制社区中仍然具有挑战性。此外，从各种故障中恢复的能力对于成功且具有韧性的实际部署至关重要，但在多技能运动研究中尚未得到充分探索。多技能框架具备在运动过程中从故障中恢复的能力（Yang等，2020），但除了小跑之外，它没有展示更多动态步态。尽管跌倒恢复已在多项先前研究中得到探讨（Hwangbo等，2019；Castano等，2019），但它是作为单一技能学习的，无法与其他技能结合使用。

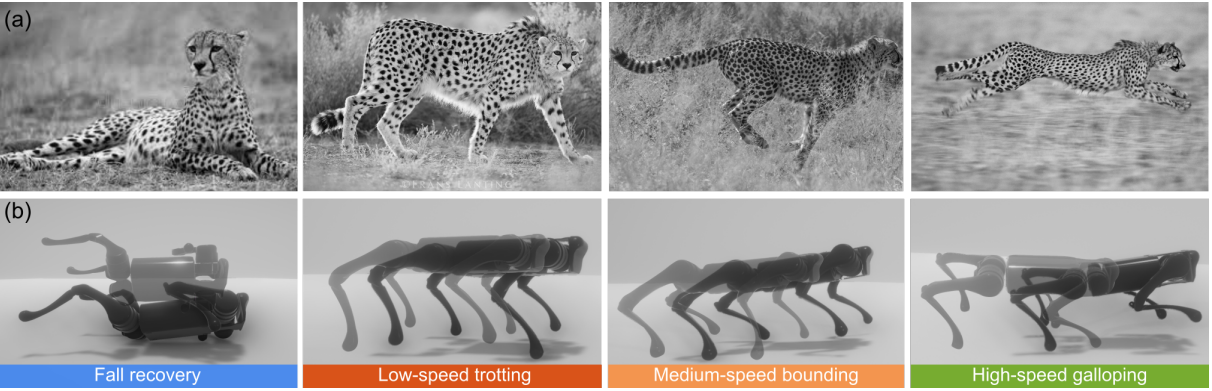


图1：四足动物和机器人根据速度增加的需求进行协调与步态转换。(a) 猎豹在速度增加时改变步态。(b) *A1* 四足机器人使用我们的多技能策略实现跌倒恢复、小跑、跳跃和疾驰技能。

复现步态模式。行为克隆或模仿学习方法已被应用于从参考运动中复现各种步态。参考运动可以从动物运动中捕捉，但其种类可能有限，也可以由模型预测控制器生成（Reske等，2021），这需要领域知识和计算效率。一般来说，这类方法难以扩展到数据集之外，缺乏所学行为的鲁棒性和多样性。已有研究尝试建模或学习参数化控制策略以实现不同风格的行走。相位引导控制器被用于在四足机器人上学习行走、小跑、踱步和跳跃之间的步态转换（Shao等，2021）。单一策略被学习用于控制具有可变摆腿、姿态和速度的各种步态（Margolis和Agrawal，2023）。然而，这类方法需要手工设计控制特定的行为参数，这需要领域知识，且切换步态不够直观。

多技能运动中的生成模型。生成模型的最新进展已实现四足机器人运动步态的低级控制。变分自编码器已被应用于学习跨运动步态的、关于摆腿高度和长度的解耦二维潜在表示（Mitchell等，2024）。给定所需的步态类型和摆动特性，该方法实现了四足机器人小跑、爬行和踱步的低级控制。此外，扩散模型已展示出通过单一策略实现多技能运动控制的能力，例如小跑、跳跃、踱步、行走和奔跑（Huang等，2024），以及行走、爬行及其转换（O’ Mahoney等，2024）。然而，这些生成模型需要专家演示，其性能依赖于数据集的质量。此外，步态类型是以控制策略的特定输入为条件的。相比之下，我们提出的分层框架覆盖高级和低级多技能运动控制，能够自主发现步态类型。

基础模型 在腿式运动中的应用。将基础模型应用于机器人学习任务

becomes an attractive approach to achieve generalized robot tasks and behaviours. Pre-trained vision-language models (VLMs) usually focus on high-level reasoning and planning to select from a set of existing low-level skills (Chen et al., 2024). However, certain low-level locomotion skills can be difficult to obtain in practice. Several attempts have also been made to apply pre-trained Large Language Models (LLMs) to achieve multiple gaits via low-level interfaces, such as foot contact patterns (Tang et al., 2023). In general, these foundation models in robotic applications require either careful prompt engineering or a huge amount of robotic data to fine-tune. In practice, the robotic data can be difficult to obtain in certain cases and the fine-tuning of large-scale models may require heavy computing resources.

Bio-inspired multi-gait locomotion. Unlike the above approaches requiring reference motions, there has been robotics research applying deep reinforcement learning approaches to acquire animal-like gait transitions according to various criteria inspired by biological principles, where reference motions are not necessary. By minimizing energy consumption, the robot can achieve gait transitions between walking, trotting and fly-trotting at different speed ranges using a single policy (Liang et al., 2024) or a hierarchical structure (Yang et al., 2022), and gait transitions from walking to trotting to bouncing (Fu et al., 2021). Another bio-inspired research modulates gait transitions according to Froude numbers (Humphreys et al., 2023). A more recent work learned gait transitions from walking to trotting on a flat ground and trotting to pronking when crossing gaps according to viability (Shafiee et al., 2024). However, galloping cannot emerge or be incorporated at high speed on mechanical robots in the above frameworks. Moreover, a series of work utilizes Central Pattern Generators (CPGs) to produce different gaits by deep reinforcement learning on quadrupedal robots. The most recent work took a coupling-driven approach to learn a policy by deep reinforcement learning to modulate the parameters of Central Pattern Generators (CPGs), where nine gaits and transitions including galloping have been produced according to Cost of Transport (CoT) (Bellegarda et al., 2024). In contrast with bio-inspired multi-gait locomotion, our proposed framework can produce galloping gait. Moreover, our approach is not constrained by biological principles as we can define customized cost terms for optimizing gait switch timing. Besides biology-inspired criteria, we can also include other cost terms, such as task-related costs.

State-of-the-art quadrupedal locomotion. A parallel line of research demonstrated impressive dynamic parkour skills on legged robots (Caluwaerts et al., 2023; Cheng et al., 2024; Zhuang et al., 2023; He et al., 2024). However, these works usually focused on navigating the robot along a series of challenging terrains and obstacles. In most cases, a simple goal-reaching task is considered in these navigation tasks, while gait patterns are not taken into account. In contrast, this paper focuses on multi-skill navigation and control tasks, i.e., reaching arbitrary goals with various gait patterns and their transitions.

Contributions. To summarize, the advantages of our proposed approach over the existing literature include: (1) we do not require reference trajectories or expert demonstrations, and can learn multi-skill locomotion purely from scratch; (2) animal-like galloping gait can be activated at high-speed locomotion; (3) autonomous fall recovery is incorporated in the multi-skill policy, enabling high robustness and requiring less human intervention; (4) flexible gait switch criteria are automatically discovered for mechanical robots. To the best of our knowledge, our work is the first multi-skill learning and optimization framework that is compatible to incorporate and synthesize multiple highly dynamic locomotion gaits (especially galloping), and produce natural, dynamically feasible transitions by automatically discovered gait switch criteria. Our work demonstrates four skills on a quadruped robot in the real world including prompt fall recovery at any stage during multi-skill locomotion. To summarize, the contributions of this paper include:

- Incorporate highly dynamic locomotion skills of bounding and galloping besides trotting into learning one coherent multi-skill policy, without the need for reference trajectories.
- Demonstrate successful trotting, bounding and galloping and their dynamically feasible and continuous transitions with one synthesized multi-skill policy on a real quadruped robot.
- Successful failure recovery at any stage of different gaits.
- Automatic discovery of gait switch criteria as motor learning progresses.

已成为实现通用机器人任务和行为的一种有吸引力的方法。预训练的视觉语言模型（VLMs）通常侧重于高级推理和规划，以从一组现有的低级技能中进行选择（Chen等，2024）。然而，某些低级运动技能在实践中可能难以获得。也有若干尝试将预训练的大语言模型（LLMs）应用于通过低级接口（如足底接触模式）实现多种步态（Tang等，2023）。总的来说，这些在机器人应用中的基础模型要么需要精心的提示工程，要么需要大量的机器人数据进行微调。在实践中，某些情况下机器人数据可能难以获取，而大规模模型的微调可能需要大量的计算资源。

仿生多步态运动。与上述需要参考运动的方法不同，已有机器人研究应用深度强化学习方法，根据受生物原理启发的各种标准来获取类似动物的步态转换，而无需参考运动。通过最小化能量消耗，机器人可以使用单一策略（Liang等人，2024）或层级结构（Yang等人，2022）在不同速度范围内实现行走、小跑和飞小跑之间的步态转换，以及从行走到小跑再到弹跳的步态转换（Fu等人，2021）。另一项仿生研究根据弗劳德数调节步态转换（Humphreys等人，2023）。最近的一项工作学习了在平坦地面上从行走到小跑的步态转换，以及根据可行性在跨越间隙时从小跑到跳跃的转换（Shafiee等人，2024）。然而，在上述框架中，疾驰无法在机械机器人上以高速涌现或被纳入。此外，一系列工作利用中枢模式发生器（CPGs）通过深度强化学习在四足机器人上产生不同的步态。最近的工作采用了一种耦合驱动方法，通过深度强化学习学习策略来调节中枢模式发生器（CPGs）的参数，其中根据运输成本（CoT）产生了包括疾驰在内的九种步态及其转换（Bellegarda等人，2024）。与仿生多步态运动相比，我们提出的框架能够产生疾驰步态。此外，我们的方法不受生物原理的约束，因为我们可以定义自定义成本项来优化步态切换时机。除了生物启发标准外，我们还可以包含其他成本项，例如任务相关成本。

最先进的四足运动。一条平行的研究路线在腿式机器人上展示了令人印象深刻的动态跑酷技能（Caluwaerts等人，2023；Cheng等人，2024；Zhuang等人，2023；He等人，2024）。然而，这些工作通常侧重于引导机器人穿越一系列具有挑战性的地形和障碍物。在大多数情况下，这些导航任务中考虑的是简单的目标到达任务，而未考虑步态模式。相比之下，本文专注于多技能导航和控制任务，即使用各种步态模式及其转换到达任意目标。

贡献。总而言之，我们提出的方法相较于现有文献的优势包括：（1）我们不需要参考轨迹或专家演示，可以完全从零开始学习多技能运动；（2）类动物疾驰步态可以在高速运动时被激活；（3）自主跌倒恢复被整合到多技能策略中，实现了高鲁棒性并减少了人工干预；（4）为机械机器人自动发现了灵活的步态切换准则。据我们所知，我们的工作第一个多技能学习与优化框架，能够兼容并整合多种高动态运动步态（尤其是疾驰），并通过自动发现的步态切换准则产生自然、动态可行的转换。我们的工作现实世界的四足机器人上展示了四种技能，包括在多技能运动过程中任何阶段的即时跌倒恢复。总而言之，本文的贡献包括：

- 将跳跃和疾驰等高度动态的运动技能与小跑一起纳入学习一个连贯多技能策略中，无需参考轨迹。
- 在真实四足机器人上，通过一个综合的多技能策略，展示成功的小跑、跳跃和疾驰及其动态可行且连续的转换。
- 在不同步态的任何阶段实现成功的故障恢复。
- 随着运动学习过程的推进，自动发现步态切换准则。

Our hierarchical multi-skill learning and optimization framework is shown in Fig. 2, which includes: (1) a set of pre-trained reusable single-skill neural network policies, each representing a single locomotion skill; (2) a task-level neural network which generates weights for each skill to produce our multiplicative composite policy; (3) the composite multi-skill policy; and (4) the outer optimization loop for the discovery of gait switch criteria represented by the relative goal distances in the horizontal plane that activate switch from trotting and bounding, and switch from bounding to galloping, respectively.

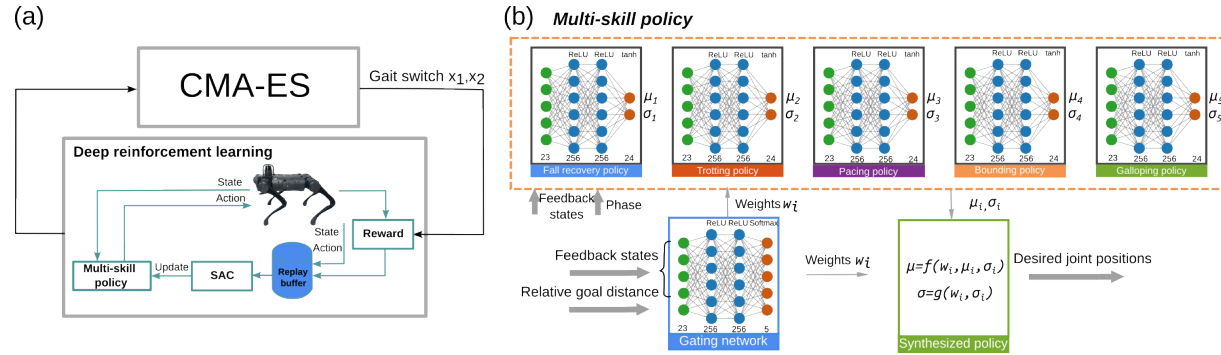


Figure 2: Proposed multi-skill learning and optimization framework. (a) Optimizing gait switch criteria in the outer loop of deep reinforcement learning. (b) Neural network architecture of a multi-skill policy. Bold arrows are the input or output outside the policy, while normal arrows are the internal input or output.

In the following sections, we will first review the details of our hierarchical learning and optimization framework in Section 2 and 3. Then we demonstrate and analyze the learned multi-skill locomotion policy on a real quadruped robot in Section 4. Finally, we conclude our work in Section 5.

2 Hierarchical multi-skill learning framework

2.1 Learning individual skills

The robot learns five individual skills separately using a systematic deep reinforcement learning framework, including fall recovery, trotting, pacing, bounding and galloping. Each locomotion skill is a feedback control policy represented by a neural network, which is learned by the Soft Actor-Critic (SAC) algorithm (Haarnoja et al., 2018). Details of the key components for our deep reinforcement learning framework are given below.

2.1.1 State observation and action space

Following the key feedback states in learning locomotion skills (Yu et al., 2023), the state input to the Actor neural network includes: (1) normalized gravity vector in the robot local frame which reflects the body orientation of the robot, (2) base angular velocity, (3) base linear velocity in the robot heading frame, and (4) joint positions. For learning periodic locomotion skills, we also included a two-dimensional phase vector ($\sin 2\pi\phi, \cos 2\pi\phi$) to represent continuous temporal information that encodes phase ϕ from 0% to 100% of a periodic motion. The actions are the desired joint positions for twelve joints, including hip roll, hip pitch and knee joints of four legs.

2.1.2 Reward design

Trotting and bounding were learned with a fixed desired velocity, while galloping was learned by maximizing velocity. The reward function for learning individual policies are composed of continuous and discrete reward

我们的分层多技能学习与优化框架如图2所示，包括：（1）一组预训练可复用单技能神经网络策略，每个策略代表一种单一运动技能；（2）一个任务级神经网络，为每种技能生成权重以构成我们的乘法复合策略；（3）复合多技能策略；（4）外部优化循环，用于发现步态切换准则，该准则由水平面上的相对目标距离表示，分别触发从小跑到跳跃的切换，以及从跳跃到疾驰的切换。

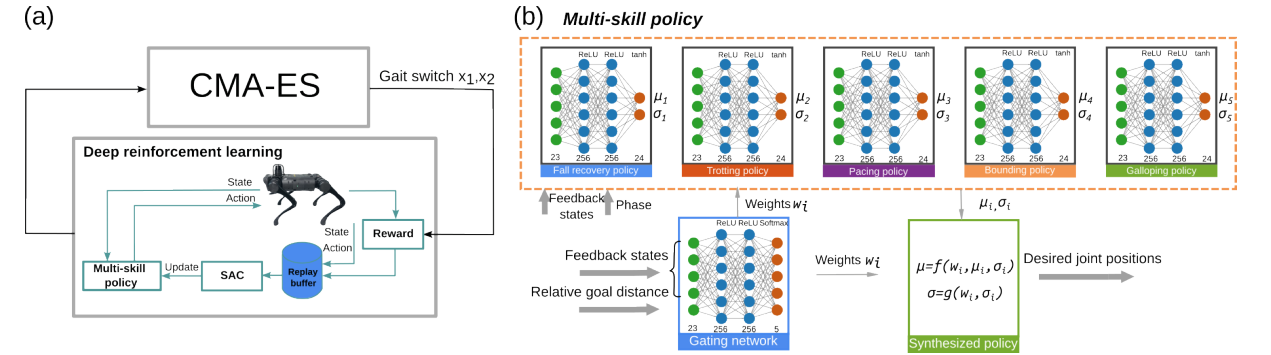


图2：提出的多技能学习与优化框架。(a) 在深度强化学习的外循环中优化步态切换准则。(b) 多技能策略的神经网络架构。粗箭头表示策略外部的输入或输出，而普通箭头表示内部的输入或输出。

在接下来的章节中，我们首先将在第2节和第3节回顾我们的分层学习与优化框架的细节。然后，我们将在第4节中展示并分析在真实四足机器人上学习到的多技能运动策略。最后，我们将在第5节中总结我们的工作。

2 分层多技能学习框架

2.1 学习个体技能

机器人使用系统的深度强化学习框架分别学习五种个体技能，包括跌倒恢复、小跑、踱步、跳跃和疾驰。每种运动技能都是由神经网络表示的反馈控制策略，通过软演员-评论家（SAC）算法（Haarnoja等人，2018）学习得到。我们深度强化学习框架的关键组成部分的详细信息如下所述。

2.1.1 状态观测与动作空间

遵循学习运动技能中的关键反馈状态（Yu等，2023），Actor神经网络的输入状态包括：（1）机器人局部坐标系中的归一化重力向量，反映机器人的机体姿态；（2）基座角速度；（3）机器人航向坐标系中的基座线速度；（4）关节位置。为学习周期性运动技能，我们还加入了二维相位向量（ $\sin 2\pi\phi, \cos 2\pi\phi$ ）来表示连续的时序信息，编码周期运动中从0%到100%的相位 ϕ 。动作是十二个关节的期望关节位置，包括四条腿的髋关节横滚、髋关节俯仰和膝关节。

2.1.2 奖励设计

小跑和跳跃是通过固定期望速度学习的，而疾驰则是通过最大化速度学习的。学习各个策略的奖励函数由连续奖励和离散奖励组成

terms. For continuous reward terms, we use a radial basis function (RBF) to formulate as follows:

$$\varphi(x, \hat{x}, \alpha) = \exp(\alpha(\hat{x} - x)^2) \quad (1)$$

where x is the continuous physical quantity, $\hat{x}$ is the corresponding reference, and α is the shape parameter which controls the width of RBF. Formulation and weight of each reward term are given in Table 1 and 2, respectively.

Table 1: Reward terms for learning quadruped locomotion skills.

Physical quantities	Reward term
Base orientation	$w_\phi \times \varphi(\phi, [0, 0, -1], -2.35)$
Base height	$w_h \times \varphi(h, \hat{h}, -51.16)$
Base linear velocity	$w_v \times \begin{cases} v_{base}^{world^2}, & \text{gallop} \\ \varphi(v_{base}^{world}, \hat{v}_{base}^{world}, -18.42), & \text{else} \end{cases}$
Joint torque	$w_\tau \times \varphi(\tau, 0, -0.004)$
Joint velocity	$w_{\dot{q}} \times \varphi(\dot{q}, 0, -0.032)$
Body ground contact	$w_{bg} \times \begin{cases} 0, & \text{base in contact with ground} \\ 1, & \text{base not in contact with ground} \end{cases}$
Foot ground contact	$w_{fg} \times \begin{cases} 0, & \text{no foot in contact with ground} \\ 1, & \text{foot in contact with ground} \end{cases}$
Symmetric foot placement	$w_{pf} \times \begin{cases} \varphi(p_{foot}^{base}, \hat{p}_{foot}^{base}, -51.16), & \text{recovery} \\ \varphi(1/4 \sum_{n=1}^4 (p_{foot,n}^{world}), p_{base}^{world}, -51.16), & \text{gaits} \end{cases}$
Swing and stance	$w_{hf} \times \varphi(h_{foot}^{world}, v_{foot}^{world}, \hat{h}_{foot}^{world}, \hat{v}_{foot}^{world}, -460.50)$
Yaw velocity	$w_{\dot{\psi}} \times \varphi(\dot{\psi}, 0, -7.47)$
Reference foot contact	$w_f \times \begin{cases} 0, & \text{not match desired foot contact} \\ 1, & \text{match desired foot contact} \end{cases}$

Table 2: Reward term weights for learning single locomotion skills.

Task	w_ϕ	w_h	w_v	w_τ	$w_{\dot{q}}$	w_{bg}	w_{fg}	w_{pf}	w_{hf}	$w_{\dot{\psi}}$	w_f
Fall recovery	0.189	0.189	0.114	0.076	0.076	0.083	0.083	0.189	0.000	0.000	0.000
Gaits	0.068	0.068	0.170	0.017	0.017	0.048	0.000	0.034	0.034	0.068	0.476

2.1.3 Reference foot contact reward

The last reward term in Table 1, i.e., reference foot contact reward, is the key to learning different gait types without reference. The desired foot contact pattern for each gait type is inspired by quadrupedal animals (Owaki and Ishiguro, 2017), as shown in Fig. 3. In this paper, we assume trotting, bounding and galloping gait as speed increases as a proof-of-concept. It should be noted that the order of the gait types is not fixed. Here we determine the gait type at different stages according to its characteristics.

2.2 Learning multi-skill locomotion

The task objective of the robot is to track an arbitrary goal in the horizontal plane with natural gait transitions from trotting to bounding to galloping. The goal is represented by (d_g, θ_g) , and the goal position in the horizontal plane is $(d_g \cos \theta_g, d_g \sin \theta_g)$. At each episode, we set a fixed goal, where $d_g \sim (0m, 15m)$ and $\theta_g \sim (-180^\circ, 180^\circ)$. The multi-skill locomotion policy is synthesized on-the-fly by multiplicative composition

项。对于c 对于连续奖励项，我们使用径向基函数（RBF）进行如下表述：

$$\varphi(x, \hat{x}, \alpha) = \exp(\alpha(\hat{x} - x)^2) \quad (1)$$

其中 x 是连续的物理量， $\hat{x}$ 是对应的参考值， α 是控制径向基函数宽度的形状参数。每个奖励项的公式和权重分别见表1和表2。

表1：用于学习四足运动技能的奖励项。

物理量	奖励项
基座姿态	$w_\phi \times \varphi(\phi, [0, 0, -1], -2.35)$
基座高度	$w_h \times \varphi(h, \hat{h}, -51.16)$
基座线速度	$w_v \times \begin{cases} v_{base}^{world^2}, & \text{疾驰} \\ \varphi(v_{base}^{world}, \hat{v}_{base}^{world}, -18.42), & \text{else} \end{cases}$
关节力矩	$w_\tau \times \varphi(\tau, 0, -0.004)$
关节速度	$w_{\dot{q}} \times \varphi(\dot{q}, 0, -0.032)$
机体地面接触	$w_{bg} \times \begin{cases} 0, & \text{基座与地面接触} \\ \emptyset, & \text{基座不接触地面} \end{cases}$
脚与地面接触	$w_{fg} \times \begin{cases} 0, & \text{无脚接触地面} \\ \emptyset, & \text{脚接触地面} \end{cases}$
对称落脚点	$w_{pf} \times \begin{cases} \varphi(p_{foot}^{base}, \hat{p}_{foot}^{base}, -51.16), & \text{恢复} \\ \varphi(1/4 \sum_{n=1}^4 (p_{foot,n}^{world}), p_{base}^{world}, -51.16), & \text{步态} \end{cases}$
摆动与支撑	$w_{hf} \times \varphi(h_{foot}^{world}, v_{foot}^{world}, \hat{h}_{foot}^{world}, \hat{v}_{foot}^{world}, -460.50)$
偏航角速度	$w_{\dot{\psi}} \times \varphi(\dot{\psi}, 0, -7.47)$
参考足部接触	$w_f \times \begin{cases} 0, & \text{不匹配期望足部接触} \\ \emptyset, & \text{匹配期望足部接触} \end{cases}$

表2：学习单一运动技能的奖励项权重。

Task	w_ϕ	w_h	w_v	w_τ	$w_{\dot{q}}$	w_{bg}	w_{fg}	w_{pf}	w_{hf}	$w_{\dot{\psi}}$	w_f
跌倒恢复	0.189	0.189	0.114	0.076	0.076	0.083	0.083	0.189	0.000	0.000	0.000
步态	0.068	0.068	0.170	0.017	0.017	0.048	0.000	0.034	0.034	0.068	0.476

2.1.3 参考足底接触奖励

表1中的最后一个奖励项，即参考足底接触奖励，是在没有参考的情况下学习不同步态类型的关键。每种步态类型的期望足部接触模式灵感来源于四足动物（Owaki和Ishiguro, 2017），如图3所示。在本文中，我们假设随着速度增加，步态依次为先跑、跳跃和疾驰，作为概念验证。需要指出的是，步态类型的顺序并非固定不变。这里我们根据各阶段的特点来确定相应的步态类型。

2.2 学习多技能运动

机器人的任务目标是在水平面上跟踪任意目标，并实现从小跑到跳跃再到疾驰的自然步态转换。目标由 (d_g, θ_g) 表示，水平面上的目标位置为 $(d_g \cos \theta_g, d_g \sin \theta_g)$ 。在每个回合中，我们设定一个固定目标，其中 $d_g \sim (0m, 15m)$ 和 $\theta_g \sim (-180^\circ, 180^\circ)$ 。多技能运动策略通过乘法组合实时合成

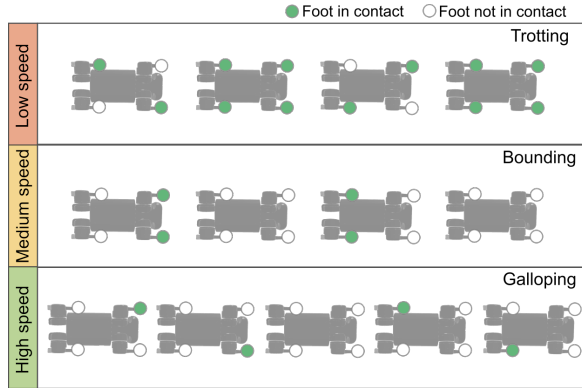


Figure 3: Foot contact patterns.

(Peng et al., 2019) of the low-level pre-trained individual policies according to the output from the high-level gating network. We will introduce more details about each component in the following sections.

2.2.1 Gating network

Our high-level gating network is a fully connected neural network with two hidden layers. Each hidden layer has 256 neurons and uses a ReLU activation function. The gating network receives the following as input: gravity vector, base angular velocity, base linear velocity, joint positions, plus normalized distance between the robot and the goal in the horizontal plane. The gating network outputs the weights for each locomotion skill which are added up to one.

2.2.2 Composite multi-skill policy

During the training of multi-skill locomotion, the parameters of the expert networks are transferred from the pre-trained single-skill policies and are fixed all the time. That is, only gating network parameters are updated by backpropagation of the gradient obtained from the designed reward. The weights for each locomotion skill generated by the gating network are then applied to synthesize the multi-skill Gaussian policy $\pi(a | s, g)$ by the multiplicative composition of n pre-trained single skills:

$$\pi(a | s, g) = \frac{1}{Z(s, g)} \prod_{i=1}^n \pi_i(a | s)^{w_i(s, g)}, \quad w_i(s, g) \geq 0 \quad (2)$$

where $\pi_i(a | s)$ is the i -th single-skill neural network policy, $w_i(s, g)$ is the weight for the corresponding skill to influence the composite policy, and $Z(s, g)$ is the normalization factor. The synthesized policy is a multiplicative composition of n Gaussian policies, i.e., n single skills. As discussed in (Peng et al., 2019), the multiplicative composition of Gaussian primitives results in another Gaussian policy, i.e., the composite policy. Due to the use of Gaussian primitives, the mean $\mu^1, \mu^2, \dots, \mu^{12}$ and standard deviation $\sigma^1, \sigma^2, \dots, \sigma^{12}$ of the synthesized Gaussian policy are obtained as follows:

$$\mu^j(s, g) = \frac{1}{\sum_{l=1}^n \frac{w_l(s, g)}{\sigma_l^j(s, g)}} \sum_{i=1}^n \frac{w_i(s, g)}{\sigma_i^j(s, g)} \mu_i^j(s, g) \quad (3)$$

$$\sigma^j(s, g) = \left(\sum_{i=1}^n \frac{w_i(s, g)}{\sigma_i^j(s, g)} \right)^{-1} \quad (4)$$

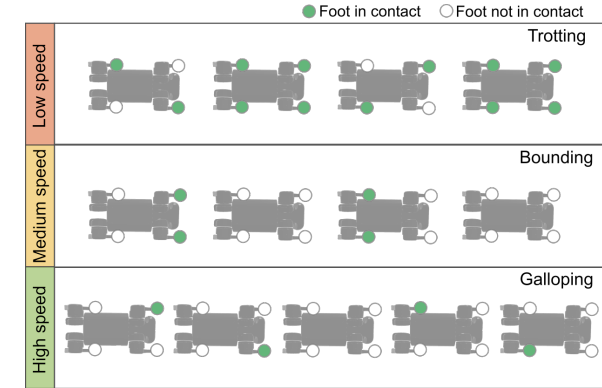


图3：足底接触模式。

(Peng等人, 2019) 根据高级门控网络的输出, 对低级预训练个体策略进行组合。我们将在以下章节中详细介绍每个组件的更多细节。

2.2.1 门控网络

我们的高级门控网络是一个具有两个隐藏层的全连接神经网络。每个隐藏层有256个神经元, 并使用ReLU激活函数。门控网络接收以下输入: 重力向量、基座角速度、基座线速度、关节位置, 以及机器人与目标在水平面上的归一化距离。门控网络输出每种运动技能的权重, 这些权重之和为1。

2.2.2 复合多技能策略

在多技能运动训练过程中, 专家网络的参数从预训练单技能策略中迁移而来, 并且始终保持固定。也就是说, 只有门控网络的参数通过从设计奖励中获得的梯度的反向传播进行更新。然后, 门控网络生成的每种运动技能的权重被应用于通过 n 预训练单一技能的乘法组合来合成多技能高斯策略 $\pi(a | s, g)$:

$$\pi(a | s, g) = \frac{1}{Z(s, g)} \prod_{i=1}^n \pi_i(a | s)^{w_i(s, g)}, \quad w_i(s, g) \geq 0 \quad (2)$$

其中 $\pi_i(a | s)$ 是第 i 个单技能神经网络策略, $w_i(s, g)$ 是对应技能影响复合策略的权重, $Z(s, g)$ 是归一化因子。合成策略是 n 个高斯策略的乘法组合, 即 n 个单一技能。如 (Peng等人, 2019) 所述, 高斯基元的乘法组合产生另一个高斯策略, 即复合策略。由于使用高斯基元, 合成高斯策略的均值 $\mu^1, \mu^2, \dots, \mu^{12}$ 和标准差 $\sigma^1, \sigma^2, \dots, \sigma^{12}$ 计算如下:

$$\mu^j(s, g) = \frac{1}{\sum_{l=1}^n \frac{w_l(s, g)}{\sigma_l^j(s, g)}} \sum_{i=1}^n \frac{w_i(s, g)}{\sigma_i^j(s, g)} \mu_i^j(s, g) \quad (3)$$

$$\sigma^j(s, g) = \left(\sum_{i=1}^n \frac{w_i(s, g)}{\sigma_i^j(s, g)} \right)^{-1} \quad (4)$$

It should be noted that we adopted a multiplicative model instead of an additive model in a hierarchical learning framework, such as Mixture of Experts (MoE) (Jacobs et al., 1991), to avoid conflicting behaviors or blending artifacts caused by the sum of primitives, as reported in (Peng et al., 2019).

2.2.3 Control framework

The pre-trained single-skill policies, the gating network, and the composite multi-skill policy run at 25 Hz. These policies together generate desired joint positions which are tracked by joint-level PD controllers at 1000 Hz. The PD controllers receive desired joint positions $\hat{q}$, measured joint positions q and joint velocities $\dot{q}$ as input and output the joint torque commands $\tau = K_p(\hat{q} - q) + K_d(0 - \dot{q})$.

2.2.4 Reward design

For multi-skill learning, we design three groups of reward terms. The most important group of reward terms is related to target-following r_g , another group is reference foot contact reward r_f , and the last group includes the rest of the reward terms in learning single locomotion skills r_e . We set the overall reward r as the weighted sum of these terms as follows:

$$r = 0.6r_g + 0.2r_f + 0.2r_e. \quad (5)$$

Goal-tracking reward: Our goal-tracking reward consists of three terms. First, the relative position reward r_{p_g} , which encourages minimization of the relative distance between the robot and the goal in the horizontal plane $d \geq 0$:

$$r_{p_g} = \varphi(p_{goal}^{world}, p_{base}^{world}, -0.74); \quad (6)$$

Second, the robot velocity reward r_{v_g} , where robot velocity towards the goal in the horizontal plane is expected to be as fast as possible:

$$r_{v_g} = v^2 \quad (7)$$

Third, the robot heading reward r_{ϕ_g} , which encourages alignment of the robot heading towards the goal:

$$r_{\phi_g} = \varphi(u_{goal,base}^{base}, [1, 0, 0], -2.35) \quad (8)$$

where $u_{goal,base}^{base}$ is the unit vector pointing from the robot base to the goal in base frame. Our full target-following reward is $r_g = r_{h_z} r_{\phi}(8r_{p_g} + 4r_{v_g} + 4r_{\phi_g})$, which increases the target-following reward weight when the robot is closer to the nominal standing pose.

Reference Foot Contact Reward: Regarding the reference foot contact reward in Table 1, learning different gaits requires different reference contact patterns. For multi-skill training, we activate different gaits according to the relative distance d between the goal and the robot as described in the target-following reward. Specifically, we use the trotting contact pattern as reference to calculate this reward if $|d| < x_1$, bounding contact pattern if $x_1 \leq |d| < x_2$, and gallop contact pattern if $|d| \geq x_2$, where x_1 and x_2 are the gait switch criteria we will automatically discover in Section 3 via an optimization loop outside the motor learning progress. Besides, it should be noted that for single skill policies, trotting has a similar velocity range as pacing and is a more common and stable gait. Therefore, when training our multi-skill policy, trotting was activated by the related reward terms when the goal is close rather than pacing. Nevertheless, technically, pacing can also be included for training a new multi-skill policy if needed.

3 Discovery of skill switching criteria

Here we propose to set up an optimization problem in the outer loop of the motor learning process to automatically discover the gait switch criteria from trotting to bounding and from bounding to galloping.

需要指出的是，我们采用乘法模型而非加法模型，在分层学习框架中，如专家混合模型 (MoE) (Jacobs 等人, 1991)，以避免如 (Peng等人, 2019) 所报道的由基元求和引起的冲突行为或混合伪影。

2.2.3 控制框架

预训练单技能策略、门控网络和复合多技能策略以25 Hz的频率运行。这些策略共同生成期望关节位置，由关节级PD控制器以1000 Hz的频率进行跟踪。PD控制器接收期望关节位置 $\hat{q}$ 、实测关节位置 q 和关节速度 $\dot{q}$ 作为输入，并输出关节力矩指令 $\tau = K_p(\hat{q} - q) + K_d(0 - \dot{q})$ 。

2.2.4 奖励设计

对于多技能学习，我们设计了三组奖励项。最重要的一组奖励项与目标跟踪 r_g 相关，另一组是参考足底接触奖励 r_f ，最后一组包括学习单一运动技能 r_e 时的其余奖励项。我们将总奖励 r 设为这些项的加权和，如下所示：

$$r = 0.6r_g + 0.2r_f + 0.2r_e. \quad (5)$$

目标跟踪奖励：我们的目标跟踪奖励由三项组成。首先，相对位置奖励 r_{p_g} ，它鼓励最小化机器人与目标在水平面 $d \geq 0$ 上的相对距离：

$$r_{p_g} = \varphi(p_{goal}^{world}, p_{base}^{world}, -0.74); \quad (6)$$

其次，机器人速度奖励 r_{v_g} ，其中期望机器人在水平面上朝向目标的速度尽可能快：

$$r_{v_g} = v^2 \quad (7)$$

第三，机器人 朝向奖励 r_{ϕ_g} ，该奖励鼓励机器人朝向与目标方向对齐 1:

$$r_{\phi_g} = \varphi(u_{goal,base}^{base}, [1, 0, 0], -2.35) \quad (8)$$

其中 $u_{goal,base}^{base}$ 是从机器人基座指向目标的单位向量，在基座坐标系中表示。我们的完整目标跟随奖励为 $r_g = r_{h_z} r_{\phi}(8r_{p_g} + 4r_{v_g} + 4r_{\phi_g})$ ，当机器人更接近标称站立姿态时，该奖励会增加目标跟随奖励的权重。

参考足底接触奖励：关于表1中的参考足底接触奖励，学习不同的步态需要不同的参考接触模式。对于多技能训练，我们根据目标跟随奖励中描述的目标与机器人之间的相对距离 d 来激活不同的步态。具体来说，如果 $|d| < x_1$ ，我们使用快步接触模式作为参考来计算该奖励；如果 $x_1 \leq |d| < x_2$ ，则使用跳跃接触模式；如果 $|d| \geq x_2$ ，则使用奔跑接触模式，其中 x_1 和 x_2 是步态切换准则，我们将在第3节中通过运动学习过程之外的优化循环自动发现这些准则。此外，需要指出的是，对于单技能策略，踱步与快步具有相似的速度范围，但快步是一种更常见且更稳定的步态。因此，在训练我们的多技能策略时，当目标较近时，相关奖励项会激活快步而非踱步。不过，从技术上讲，如果需要，踱步也可以纳入新多技能策略的训练中。

3 发现技能切换准则

在此，我们提出在运动学习过程的外循环中建立一个优化问题，以自动发现从小跑到跳跃、从跳跃到疾驰的步态切换准则。

We use covariance matrix adaptation evolution strategy (CMA-ES) (Hansen, 2016), which is a derivative-free evolution strategies inspired by biological evolution for optimization. We aim to find gait switch criteria x_1m, x_2m to maximize the sum of target following reward r_g over each episode. The optimization problem is formulated as:

$$\begin{aligned} \arg \min_{x_1, x_2} \sum_{i=1}^N -r_g(x_1, x_2) \\ \text{s.t.} \quad & 0 \leq x_1 \leq 15, \\ & 0 \leq x_2 \leq 15, \\ & x_1 < x_2. \end{aligned}$$

where x_1, x_2 are the decision variables representing the relative distance between robot and goal in the horizontal plane to activate the switch between trotting and bounding, bounding and galloping gaits, respectively. N is the number of time steps in an episode.

4 Experimental results

This section first introduces experimental setup and then presents the optimization process of finding the proper gait switch criteria. We then demonstrate that the proposed multi-skill policy can achieve versatile locomotion skills by giving normalized goal distance input. Moreover, we show that the robot can recover from failures at any time during multi-skill locomotion. Furthermore, we showcase our proposed approach outperforms the baseline, i.e., manual switch between single skills, in terms of stability, smoothness and robustness.

4.1 Experimental setup

Multi-skill training setup. We sample 5000 steps from the composite multi-skill policy for each training epoch, i.e., 20 episodes without early termination; Each episode lasts 10 seconds; Batch size is 128; Replay buffer size is 1e6; Learning rate is 3e-4; Weight decay is 1e-6; Soft target update is 0.001; Discount factor is 0.995 for fall recovery and 0.955 for locomotion gaits.

Skill switch criteria optimization setup. We set initial gait switch criteria as $x_1 = 2.0m$ and $x_2 = 5.0m$ to warmstart the optimization, population size as 50, and $\sigma = 1.0m$. The CMA-ES optimization runs for an iteration for every 20 iterations of inner loop deep reinforcement learning.

Goal trajectories setup. We give normalized relative goal distance in the x- and y-axis in the robot heading frame via a joystick in real-world tests to encourage the emergence of multiple dynamic skills and their transitions. An example goal trajectory is shown in Fig. 4.

Velocity estimation. For the deployment of multi-skill policies learned in simulation on real robots, sensing errors and uncertainties usually cause the discrepancies between simulation and the real world. Unlike other state observations we selected for learning, base linear velocity cannot be obtained directly and needs to be estimated via leg kinematics or visual odometry, where the estimation results perform bad during foot slipping or highly dynamic motions (Ji et al., 2022). To this end, similar to (Ji et al., 2022), we train a separate velocity estimator, to obtain the estimation of unavailable or unreliable states given the sensory information of more reliable states. The input to the state estimator is 66-dimensional including gravity vector from roll and pitch measurements from IMU, two-step history of gravity vector, base angular velocity from IMU, two-step history of base angular velocity, joint positions, two-step history of joint positions, and joint velocity from motor encoders. The output is 3-dimensional estimated base linear velocity. The estimator network is composed of two hidden layers each has 256 neurons and uses a ReLU activation function. After we obtain the locomotion policies in simulation, we collect 215000 pairs of input-output for training the

我们采用协方差矩阵自适应进化策略（CMA-ES）（Hansen, 2016），这是一种受生物进化启发的无导数进化策略，用于优化。我们的目标是找到步态切换准则 x_1m, x_2m ，以最大化每个回合中目标跟踪奖励 r_g 的总和。优化问题表述如下：

$$\begin{aligned} \arg \min_{x_1, x_2} \sum_{i=1}^N -r_g(x_1, x_2) \\ \text{s.t.} \quad & 0 \leq x_1 \leq 15, \\ & 0 \leq x_2 \leq 15, \\ & x_1 < x_2. \end{aligned}$$

其中 x_1, x_2 是决策变量，分别表示机器人与目标在水平面上的相对距离，用于激活小跑与跳跃、跳跃与疾驰步态之间的切换。 N 是一个回合中的时间步数。

4 实验结果

本节首先介绍实验设置，然后展示寻找合适步态切换准则的优化过程。接着，我们通过给定归一化目标距离输入，证明所提出的多技能策略能够实现多功能运动技能。此外，我们展示了机器人在多技能运动过程中能够随时从失败中恢复。最后，我们展示了所提出的方法在稳定性、平滑性和鲁棒性方面优于基线方法，即在单一技能之间进行手动切换。

4.1 实验设置

多技能训练设置。 每个训练轮次从复合多技能策略中采样5000步，即20个回合且无提前终止；每个回合持续10秒；批大小为128；回放缓冲区大小为1e6；学习率为3e-4；权重衰减为1e-6；软目标更新为0.001；折扣因子在跌倒恢复中为0.995，在运动步态中为0.955。

技能切换准则优化设置。 我们设置初始步态切换准则为 $x_1 = 2.0m$ 和 $x_2 = 5.0m$ 以热启动优化，种群规模为50，以及 $\sigma = 1.0m$ 。CMA-ES优化每进行20次内循环深度强化学习迭代，便运行一次迭代。

目标轨迹设置。 在真实世界测试中，我们通过操纵杆在机器人航向坐标系中给出x轴和y轴上的归一化相对目标距离，以促进多种动态技能及其转换的出现。图4展示了一个示例目标轨迹。

速度估计。 为了将在仿真中学习到的多技能策略部署到真实机器人上，感知误差和不确定性通常会导致仿真与现实世界之间的差异。与我们为学习而选择的其他状态观测不同，基座线速度无法直接获得，需要通过腿部运动学或视觉里程计进行估计，而在脚部打滑或高度动态运动期间，估计结果表现不佳（Ji等人，2022）。为此，类似于（Ji等人，2022），我们训练了一个单独的速度估计器，以便在给定更可靠状态的感知信息的情况下，估计不可用或不可靠的状态。状态估计器的输入为66维，包括来自IMU的俯仰角和横滚角测量得到的重力向量、两步历史的重力向量、来自IMU的基座角速度、两步历史的基座角速度、关节位置、两步历史的关节位置以及来自电机编码器的关节速度。输出为3维的估计的基座线速度。估计器网络由两个隐藏层组成，每层有256个神经元，并使用ReLU激活函数。在仿真中获得运动策略后，我们收集了215000对输入-输出数据用于训练

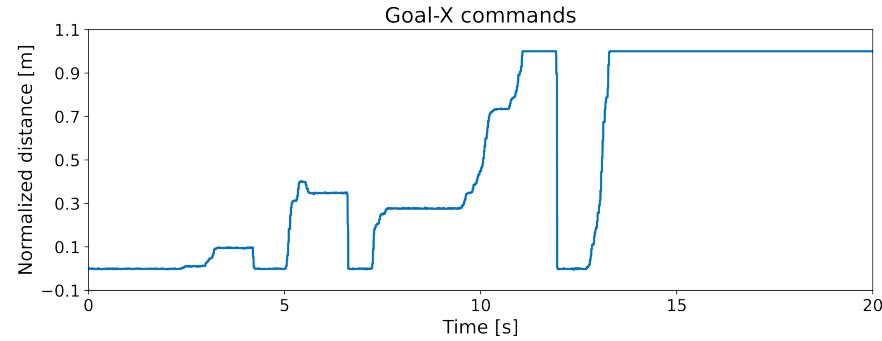


Figure 4: Normalized relative goal command in robot heading frame is given to encourage fall recovery, trotting, bounding, galloping and their transitions.

estimator network via supervised learning. We use mean squared error loss (ground truth velocity versus velocity estimation generated by the neural network) for training, where learning rate is 0.001, weight decay is 0.0005, and batch size is 1024.

4.2 Optimized skill switch criteria

From Fig. 5, we can see that the best cost value keeps decreasing and reaches a local minimum value after 27 iterations of optimization, where the corresponding trot-bound and bound-gallop switch criteria are $x_1 = 2.2m$ and $x_2 = 4.3m$, respectively. It should be noted that the optimized gait switch criteria are not exactly where the gait transitions occur in practice since they are only incorporated in reward functions. Instead, the gait transitions during multi-skill locomotion is naturally learned via the optimized gait switch criteria.

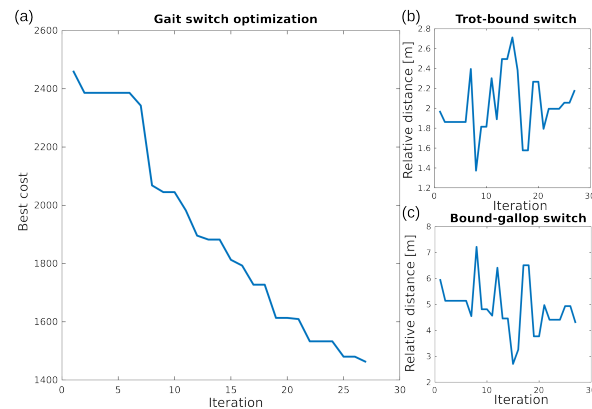


Figure 5: Results of CMA-ES optimization for gait switch criteria in learning multi-skill locomotion. (a) Best cost during CMA-ES optimization. (b) Optimized relative distance for switching from trotting to bounding. (c) Optimized relative distance for switching from bounding to galloping.

4.3 Multi-skill locomotion with continuous skill transitions

With the learned multi-skill locomotion policy, the robot is able to demonstrate trotting, bounding, galloping, and prompt fall recovery whenever necessary, as shown in Fig. 6(a) and the accompanying video. The corresponding goal commands given via joystick is shown in Fig. 4. Here we only report the experimental results in the real world. Refer to the accompanying video for more simulation and robustness tests.

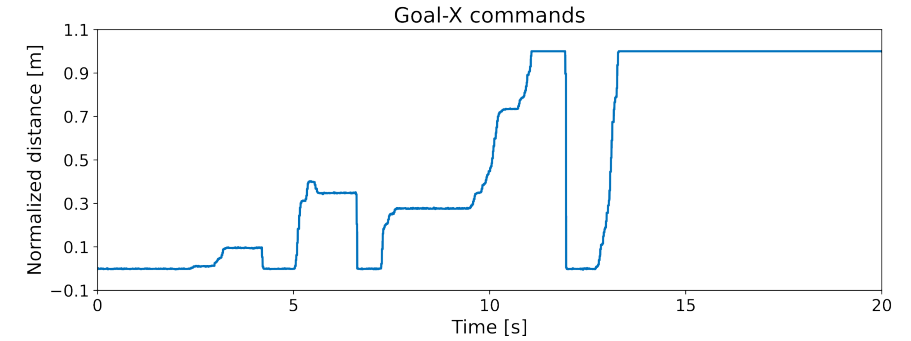


图4：在机器人航向坐标系中给出归一化相对目标指令，以促进跌倒恢复、小跑、跳跃、疾驰及其转换。

估计器网络，采用监督学习。我们使用均方误差损失（真实速度与神经网络生成的速度估计之间的差异）进行训练，其中学习率为0.001，权重衰减为0.0005，批大小为1024。

4.2 优化技能切换准则

从图5可以看出，最佳成本值持续下降，并在27次优化迭代后达到局部最小值，此时对应的小跑-跳跃和跳跃-疾驰切换准则分别为 $x_1 = 2.2m$ 和 $x_2 = 4.3m$ 。需要注意的是，优化后的步态切换准则并不完全等同于实际步态转换发生的位置，因为它们仅被纳入奖励函数中。相反，多技能运动过程中的步态转换是通过优化后的步态切换准则自然学习得到的。

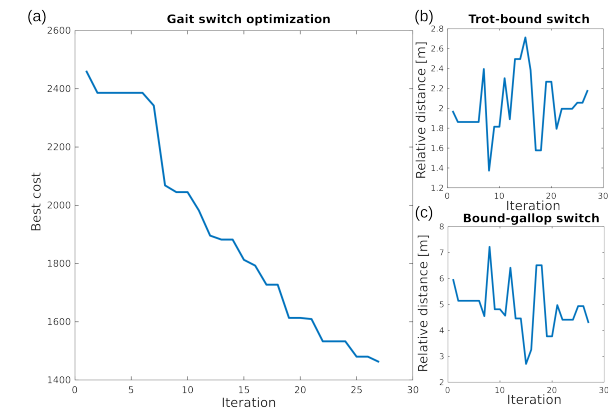


图5：多技能运动学习中步态切换准则的CMA-ES优化结果。(a) CMA-ES优化期间的最佳成本。(b) 从小跑切换到跳跃的优化相对距离。(c) 从跳跃切换到疾驰的优化相对距离。

4.3 多技能运动与连续技能转换

借助学习到的多技能运动策略，机器人能够展示小跑、跳跃、疾驰以及必要时迅速跌倒恢复的能力，如图6(a)及随附视频所示。通过操纵杆给出的相应目标指令如图4所示。此处仅报告现实世界中的实验结果。更多仿真和鲁棒性测试请参阅随附视频。

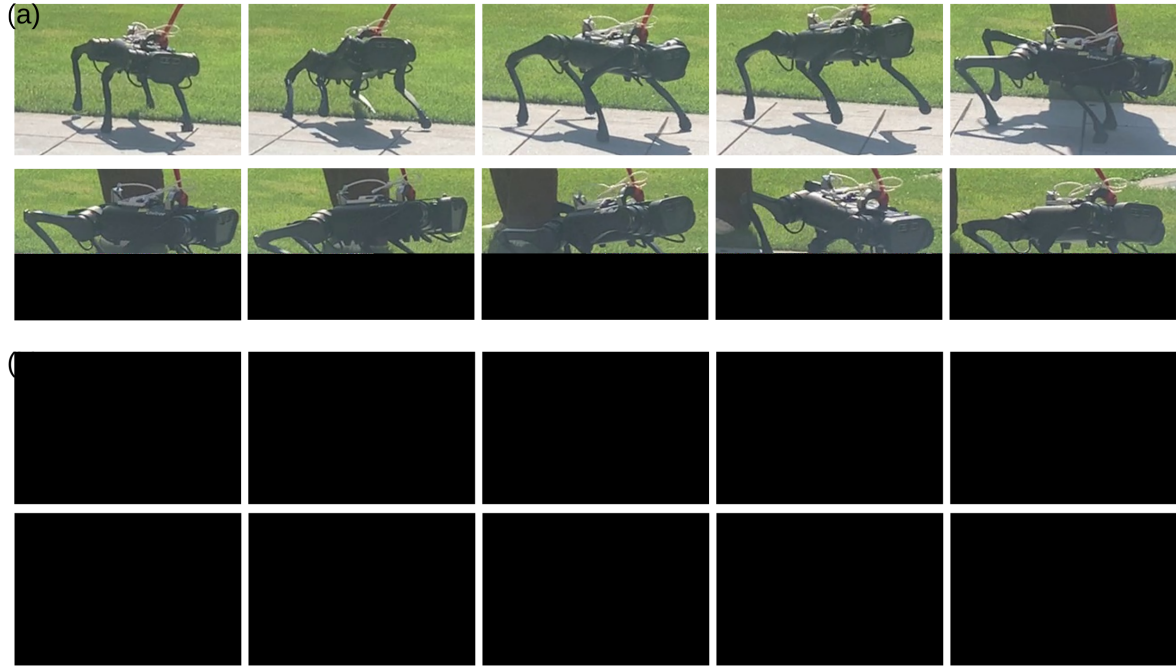


Figure 6: Comparison of multi-skill locomotion using our proposed approach and baseline. (a) Learned multi-skill locomotion by following goal trajectories in Fig. 4. (b) Baseline approach by manually switching between learned single skills. The robot failed after a discrete switch from bounding to galloping.

Estimated speed. We show the estimated speed in the horizontal plane of the robot heading frame for 20 seconds of the multi-skill locomotion, see Fig. 7(a). The robot shows an increasing velocity for trotting, bounding and galloping skills.

Expert weights. Figure 7(b) shows the weights for each single-skill policy generated by the gating network correspondingly. The fall recovery expert dominates during the recovery motion. For the three locomotion gaits exhibited, each corresponding expert has the largest weight among all. However, compared to bounding and galloping gaits where the corresponding expert dominates, the trotting expert is co-acting more together with other experts during trotting in contributing to the synthesized policy. Furthermore, to have a detailed view of the influence of each expert in different quadruped locomotion skills, we visualized the weight for each expert of the four demonstrated skills at different time steps. From Fig. 8, we clearly see the composition of single-skill policies for each motion, and each motion utilizes all the five single skills.

Euler angles. The corresponding roll and pitch angles are shown in Fig. 9. When the robot encountered falls, the roll and pitch angles could become large at first and return to normal range during fall recovery. In other case, these two euler angles have clear cyclic pattern. Moreover, from trotting to bounding to galloping, the magnitude of the euler angles becomes larger, indicating the motion becomes more dynamic.

4.4 Ablation studies

We compare our proposed multi-skill learning and optimization approach with baseline approach, i.e., manual switch between different skill primitives. For the single skills, trotting and bounding were learned with a fixed desired velocity, while galloping was learned by maximizing velocity. After multi-skill learning with the parameters of each expert network fixed, trotting motion is synthesized by the gating network at a lower speed range, and galloping motion is synthesized in a more dynamically feasible pattern with a higher success rate of real-world deployment. See the snapshots in Fig. 6 and the accompanying video for more details.

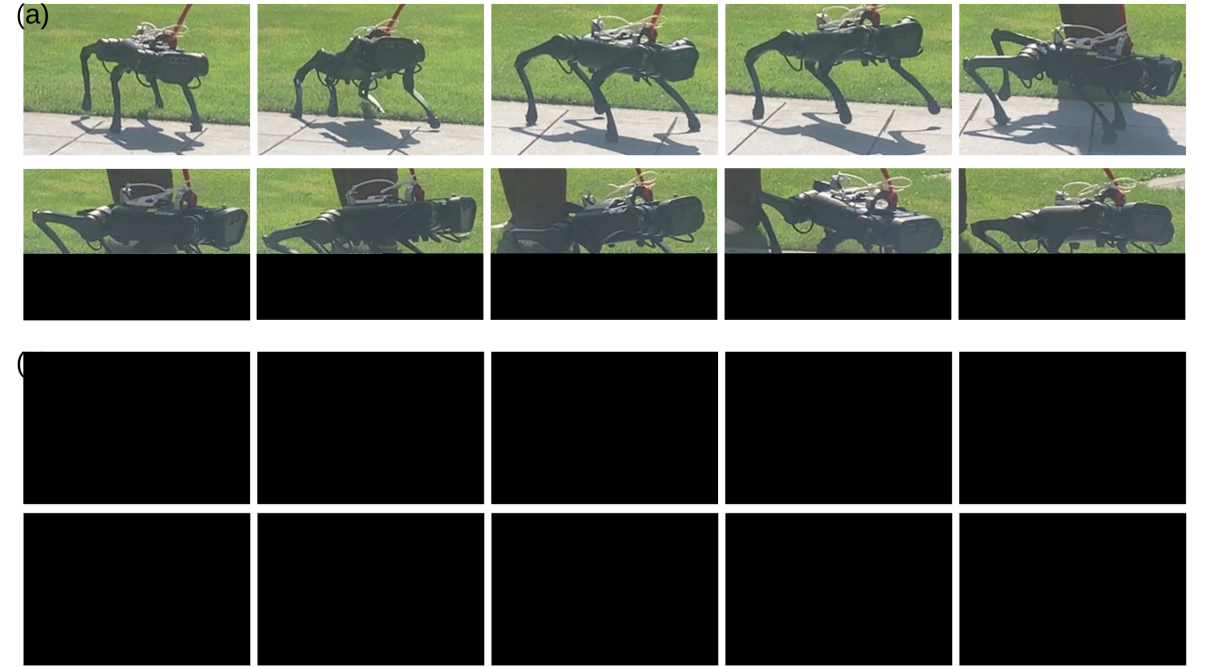


图6: 使用我们提出的方法与基线方法进行多技能运动的对比。(a) 通过遵循图4中的目标轨迹学习到的多技能运动。(b) 通过在已学习的单一技能之间手动切换的基线方法。机器人在从跳跃到疾驰的离散切换后失败。

估计速度。我们展示了多技能运动20秒内机器人航向坐标系水平面上的估计速度，见图7(a)。机器人在小跑、跳跃和疾驰技能中表现出递增的速度。

专家权重。图7(b)显示了门控网络对应生成的每个单一技能策略的权重。跌倒恢复专家在恢复运动期间占据主导地位。对于所展示的三种运动步态，每个对应的专家在所有专家中拥有最大的权重。然而，与跳跃和疾驰步态中对应专家占据主导地位相比，小跑专家在小跑期间与其他专家更多地协同作用，共同为合成策略做出贡献。此外，为了详细了解每个专家在不同四足运动技能中的影响，我们可视化了四种演示技能中每个专家在不同时间步的权重。从图8中，我们可以清晰地看到每种运动所对应的单一技能策略的组成，并且每种运动都利用了全部五个单一技能。

欧拉角。相应的横滚角和俯仰角如图9所示。当机器人遭遇跌倒时，横滚角和俯仰角起初可能变得较大，并在跌倒恢复过程中回到正常范围。在其他情况下，这两个欧拉角呈现出明显的循环模式。此外，从小跑到跳跃再到疾驰，欧拉角的幅值逐渐增大，表明运动变得更加动态。

4.4 消融研究

我们将所提出的多技能学习与优化方法与基线方法进行了比较，即在不同的技能原语之间进行手动切换。对于单一技能，小跑和跳跃是在固定期望速度下学习的，而疾驰则是通过最大化速度来学习的。在固定每个专家网络参数的情况下进行多技能学习后，门控网络在较低速度范围内合成小跑运动，并以更高的实际部署成功率合成更具动态可行性的疾驰运动。更多细节请参见图6中的快照和随附视频。



Figure 7: Versatile locomotion with continuous gait transitions. (a) Estimated horizontal speed of the robot during multi-skill locomotion, and (b) Expert weights showing each motion utilizes all the five experts with one related expert dominates.

For the baseline approach, the robot failed when manually switching from bounding to galloping, sometimes causing automatic shutdown due to the power protection for Unitree robots. In the case when it failed without triggering the power protection, we can manually activate fall recovery skill, and the robot can then recover from failure to standing still. However, to resume locomotion, it requires another discrete switch from standing to trotting, bringing further instability issues. In contrast, our multi-skill policy can directly transition from failure to trotting without the intermediate phase in a dynamic fashion. When discretely switching from trotting to bounding at an improper gait phase, the knee joints of the rear legs may become very close to the ground or the front legs lift very high in the following several time steps. As shown in Fig. 10, manual switch caused dynamically instability, such as abrupt changes in estimated velocity. While our approach enables a smoother and continuous gait transitions in the real-world deployment.



图7：具有连续步态转换的多功能运动。(a) 机器人在多技能运动过程中的估计水平速度，以及(b) 专家权重显示每个动作都利用了全部五个专家，其中一个相关专家占主导地位。

对于基线方法，机器人在从跳跃手动切换到疾驰时失败，有时会因宇树机器人的电源保护而触发自动关机。在未触发电源保护而失败的情况下，我们可以手动激活跌倒恢复技能，机器人随后能够从失败状态恢复到静止站立。然而，要恢复运动，还需要从站立到小跑进行另一次离散切换，这会带来进一步的不稳定问题。相比之下，我们的多技能策略可以直接从失败状态动态地过渡到小跑，无需中间阶段。当在不合适的步态相位从小跑离散切换到跳跃时，后腿的膝关节在接下来的几个时间步内可能会非常接近地面，或者前腿抬得很高。如图10所示，手动切换导致了动态不稳定，例如估计速度的突变。而我们的方法在实际部署中实现了更平滑、连续的步态转换。

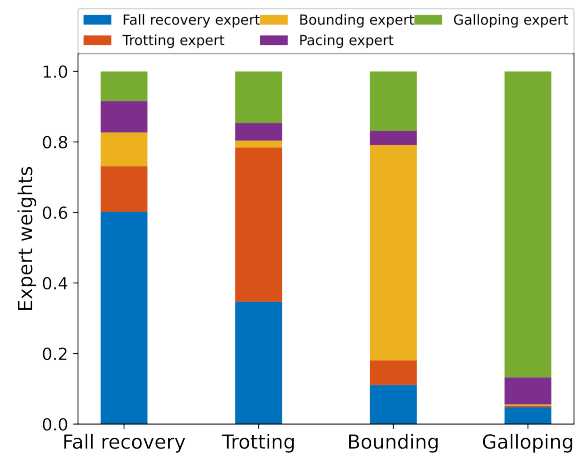


Figure 8: Composition of five skill primitives for fall recovery, trotting, bounding and galloping during multi-skill locomotion at 0.2s, 2.5s, 8.0s and 18.0s, respectively.

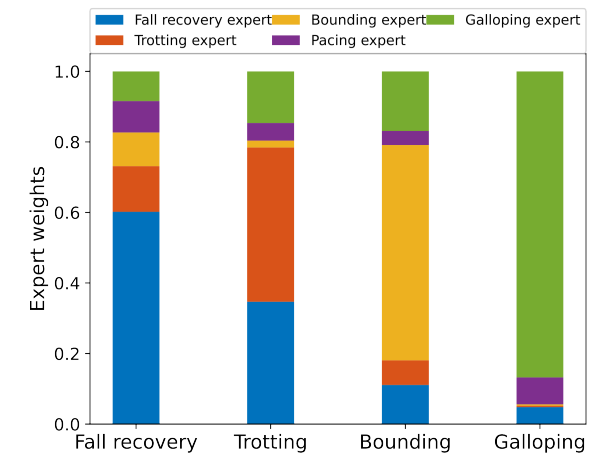


图8: 多技能运动中, 分别在0.2秒、2.5秒、8.0秒和18.0秒时, 跌倒恢复、小跑、跳跃和疾驰五种技能原语的组合。

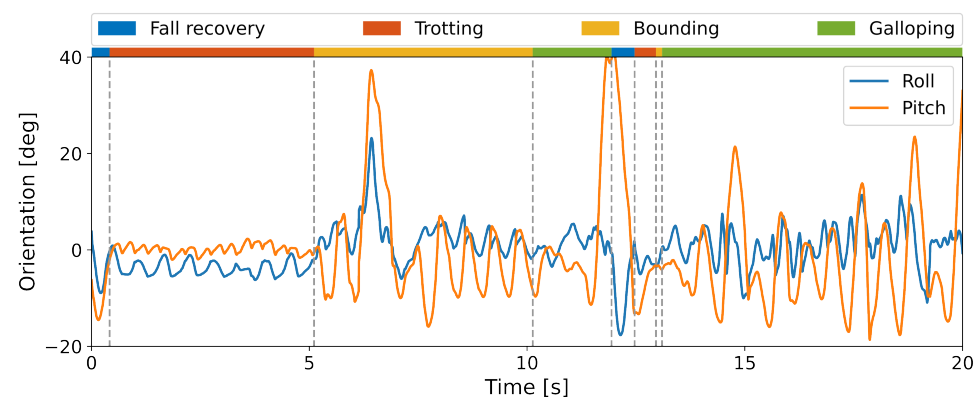


Figure 9: Roll and pitch angles during multi-skill locomotion.

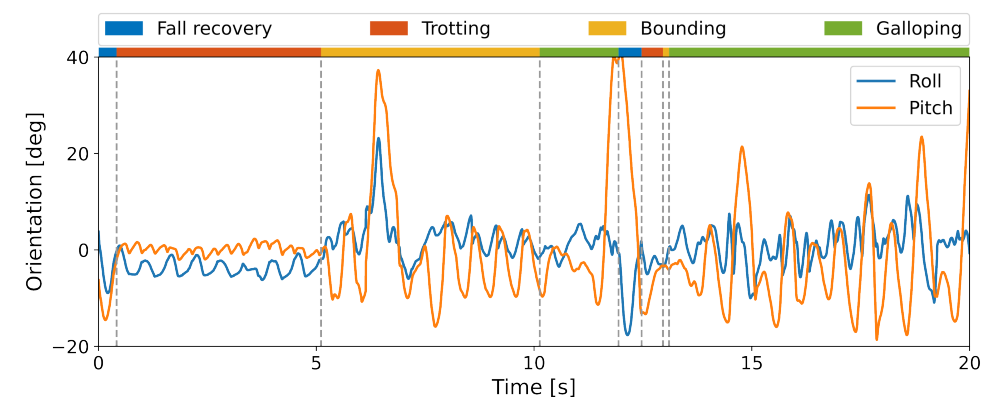


图9: 多技能运动过程中的横滚角和俯仰角。

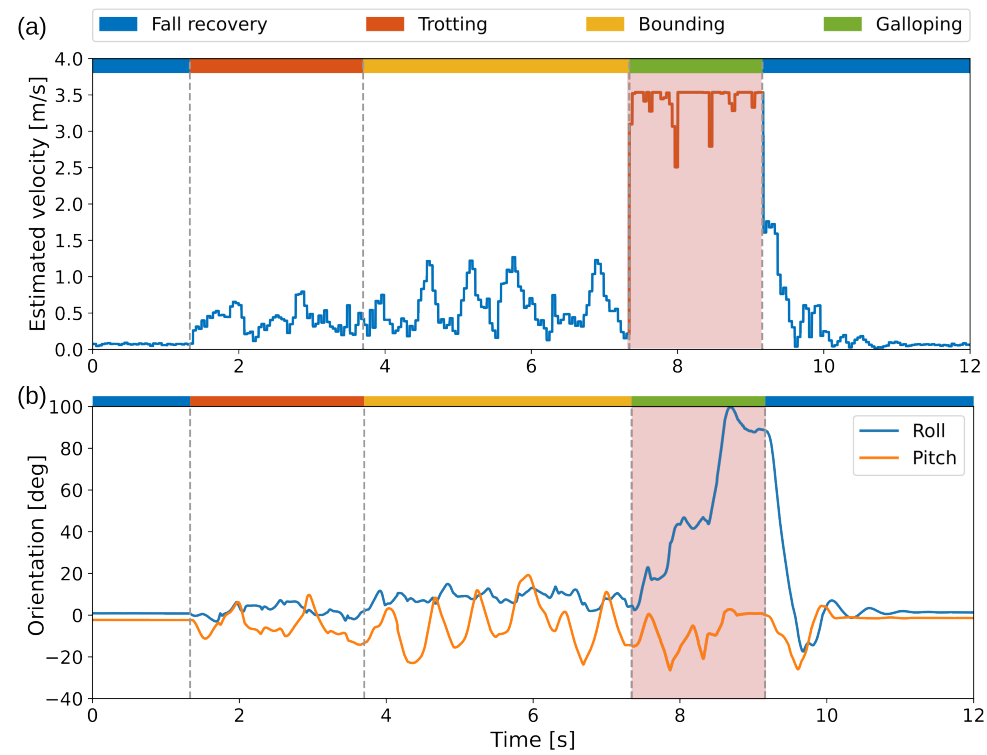


Figure 10: Performance of baseline approach by manually switching from fall recovery to trotting to bounding. Robot failed to switch to galloping from bounding (red shaded areas), but was then able to perform a successful recovery from failure to standing still. (a) Estimated horizontal speed. (b) Roll and pitch angles.

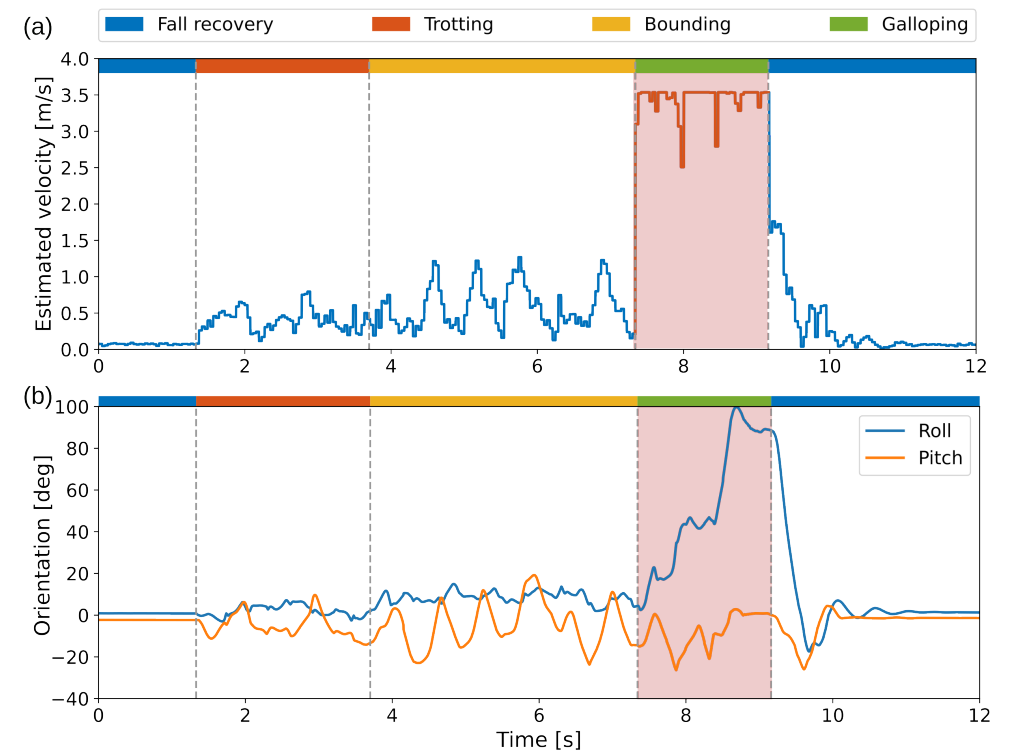


图10：通过手动从跌倒恢复切换到小跑再到跳跃的基线方法性能。机器人未能从跳跃切换到疾驰（红色阴影区域），但随后能够从失败中成功恢复到静止站立。（a）估计的水平速度。（b）横滚角和俯仰角。

5 Conclusions

This research developed a hierarchical learning and optimization framework to achieve multi-skill locomotion with optimized gait switch criteria without the need for any reference trajectories or expert demonstrations. The robot demonstrates continuous gait transitions among trotting, bounding and galloping skills as locomotion speed increases. Our learned multi-skill policy can also incorporate the fall recovery skill which enables the robot to recover promptly and resume locomotion whenever it becomes unstable or falls during different gaits. Thus, the robot requires less human intervention to operate autonomously in a remote working space, enabling versatile applications. Compared with the existing end-to-end learning framework using a single policy, our hierarchical framework is bio-inspired and more efficient in fine-tuning for various tasks since it does not require learning from scratch to adapt to new tasks. Moreover, by optimizing gait switch criteria as motor learning progresses, we avoid manually specifying the criteria with biased human knowledge distillation. The formulation can be easily adapted and generalized to different tasks or scenarios by customizing the cost function.

One limitation of our approach is that we found sim-to-real discrepancies still exist in galloping motion. In simulation, the galloping is more stable without any failures. In the real world, we cannot ensure 100 % success rate of galloping for very long periods. This can be due to various reasons. Compared to other locomotion skills, galloping is inherently a very unstable locomotion skill, since only one foot is in contact with the ground at one time. Slight sim-to-real discrepancies can cause huge difference and even failures, such as deformable foot pads on Unitree A1 robot, ground friction, velocity estimation for out-of-the-distribution motion. Another reason is the goal command we give in real-world tests is different from the simulation. For training in simulation, we give goal position directly in the world frame, while in the real world, due to the lack of body position feedback, we give normalized relative goal distance in robot heading frame via joystick, which is not possible to reproduce the same goal commands as in simulation via joystick.

For future work, we plan to further resolve the sim-to-real gap in galloping motion. We would also like to investigate the scalability of the proposed framework to more skills and more complex tasks as it would need more reward engineering. Furthermore, since our proposed approach can generate multi-skill locomotion data without any reference, one interesting application of our proposed framework would be preparing datasets for the training and fine-tuning of the generalist polices for legged robots, such as diffusion models or OpenVLA (Kim et al., 2024).

Acknowledgments

This work was supported by the UKRI Future Leaders Fellowship [MR/V025333/1] (RoboHike). Fernando Acero is supported by the UKRI CDT in Foundational Artificial Intelligence (EP/S021566/1).

5 结论

本研究开发了一种分层学习与优化框架，以实现具有优化步态切换准则的多技能运动，无需任何参考轨迹或专家演示。随着运动速度的增加，机器人展示了在小跑、跳跃和疾驰技能之间的连续步态转换。我们学到的多技能策略还可以整合跌倒恢复技能，使机器人在不同步态下变得不稳定或跌倒时能够迅速恢复并继续运动。因此，机器人在远程工作空间中自主运行所需的人工干预更少，从而实现多种应用。与使用单一策略的现有端到端学习框架相比，我们的分层框架具有仿生特性，并且在针对各种任务进行微调时更加高效，因为它不需要从零开始学习以适应新任务。此外，通过在运动学习过程中优化步态切换准则，我们避免了使用带有偏见的人类知识提炼来手动指定准则。该公式可以通过定制代价函数轻松适应并推广到不同的任务或场景。

我们方法的一个局限性在于，我们发现疾驰运动中仍然存在仿真到现实的差异。在仿真中，疾驰更加稳定，不会出现任何失败。在现实世界中，我们无法确保长时间疾驰的100%成功率。这可能是由多种原因造成的。与其他运动技能相比，疾驰本质上是一种非常不稳定的运动技能，因为同一时间只有一只脚与地面接触。轻微的仿真到现实差异可能导致巨大偏差甚至失败，例如宇树A1机器人上的可变形脚垫、地面摩擦力、分布外运动的速度估计等。另一个原因是我们在真实世界测试中给出的目标指令与仿真中的不同。在仿真训练中，我们直接在世界坐标系中给出目标位置，而在现实世界中，由于缺乏身体位置反馈，我们通过操纵杆在机器人航向坐标系中给出归一化相对目标距离，这使得无法通过操纵杆复现与仿真中相同的目标指令。

在未来的工作中，我们计划进一步解决疾驰运动中的仿真到现实的差距。我们还希望研究所提出框架在更多技能和更复杂任务上的可扩展性，因为这需要更多的奖励工程。此外，由于我们提出的方法无需任何参考即可生成多技能运动数据，我们框架的一个有趣应用是为腿式机器人的通用策略（如扩散模型或OpenVLA（Kim等人，2024））的训练和微调准备数据集。

致谢

这项工作得到了UKRI未来领袖奖学金 [MR/V025333/1] (RoboHike)的支持。Fernando Acero得到了UKRI基础人工智能博士培训中心(EP/S021566/1)的支持。

References

Bellegarda, G., Shafiee, M., and Ijspeert, A. (2024). Allgaits: Learning all quadruped gaits and transitions. *arXiv preprint arXiv:2411.04787*.

Caluwaerts, K., Iscen, A., Kew, J. C., Yu, W., Zhang, T., Freeman, D., Lee, K.-H., Lee, L., Saliceti, S., Zhuang, V., et al. (2023). Barkour: Benchmarking animal-level agility with quadruped robots. *arXiv preprint arXiv:2305.14654*.

Castano, J. A., Zhou, C., and Tsagarakis, N. (2019). Design a fall recovery strategy for a wheel-legged quadruped robot using stability feature space. In *2019 IEEE International Conference on Robotics and Biomimetics (ROBIO)*, pages 41–46. IEEE.

Chen, A. S., Lessing, A. M., Tang, A., Chada, G., Smith, L., Levine, S., and Finn, C. (2024). Commonsense reasoning for legged robot adaptation with vision-language models. *arXiv preprint arXiv:2407.02666*.

Cheng, X., Shi, K., Agarwal, A., and Pathak, D. (2024). Extreme parkour with legged robots. In *International Conference*.

Fu, Z., Kumar, A., Malik, J., and Pathak, D. (2021). Minimizing energy consumption leads to the emergence of gaits in legged robots. In *Conference on Robot Learning*. PMLR.

Haarnoja, T., Zhou, A., Abbeel, P., and Levine, S. (2018). Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. In *International Conference on Machine Learning*, pages 1861–1870. PMLR.

Hansen, N. (2016). The cma evolution strategy: A tutorial. *arXiv preprint arXiv:1604.00772*.

He, T., Zhang, C., Xiao, W., He, G., Liu, C., and Shi, G. (2024). Agile but safe: Learning collision-free high-speed legged locomotion. In *Robotics: Science and Systems*.

Hoyt, D. F. and Taylor, C. R. (1981). Gait and the energetics of locomotion in horses. *Nature*, 292(5820):239–240.

Huang, X., Chi, Y., Wang, R., Li, Z., Peng, X. B., Shao, S., Nikolic, B., and Sreenath, K. (2024). Diffuse-loco: Real-time legged locomotion control with diffusion from offline datasets. In *Conference on Robot Learning*.

Humphreys, J., Li, J., Wan, Y., Gao, H., and Zhou, C. (2023). Bio-inspired gait transitions for quadruped locomotion. *IEEE Robotics and Automation Letters*.

Hwangbo, J., Lee, J., Dosovitskiy, A., Bellicoso, D., Tsounis, V., Koltun, V., and Hutter, M. (2019). Learning agile and dynamic motor skills for legged robots. *Science Robotics*, 4(26):eaau5872.

Jacobs, R. A., Jordan, M. I., Nowlan, S. J., and Hinton, G. E. (1991). Adaptive mixtures of local experts. *Neural computation*, 3(1):79–87.

Ji, G., Mun, J., Kim, H., and Hwangbo, J. (2022). Concurrent training of a control policy and a state estimator for dynamic and robust legged locomotion. *IEEE Robotics and Automation Letters*, 7(2):4630–4637.

Kim, M. J., Pertsch, K., Karamcheti, S., Xiao, T., Balakrishna, A., Nair, S., Rafailov, R., Foster, E., Lam, G., Sanketi, P., et al. (2024). Openvla: An open-source vision-language-action model. *arXiv preprint arXiv:2406.09246*.

Liang, B., Sun, L., Zhu, X., Zhang, B., Xiong, Z., Li, C., Sreenath, K., and Tomizuka, M. (2024). Adaptive energy regularization for autonomous gait transition and energy-efficient quadruped locomotion. *arXiv preprint arXiv:2403.20001*.

参考文献

Bellegarda, G., Shafiee, M., 和 Ijspeert, A. (2024). Allgaits: 学习所有四足步态及其转换.*arXiv*预印本 *arXiv:2411.04787*. Caluwaerts, K., Iscen, A., Kew, J. C., Yu, W., Zhang, T., Freeman, D., Lee, K.-H., Lee, L., Saliceti, S., Zhuang, V., 等. (2023). Barkour: 以四足机器人基准测试动物级敏捷性. *arXiv*预印本 *arXiv:2305.14654*. Castano, J. A., Zhou, C., 和 Tsagarakis, N. (2019). 利用稳定性特征空间为轮腿式四足机器人设计跌倒恢复策略. 在2019年*IEEE*机器人与仿生学国际会议(*ROBIO*)中, 第41–46页. IEEE. Chen, A. S., Lessing, A. M., Tang, A., Chada, G., Smith, L., Levine, S., 和 Finn, C. (2024). 利用视觉语言模型进行腿式机器人适应的常识推理. *arXiv*预印本 *arXiv:2407.02666*. Cheng, X., Shi, K., Agarwal, A., 和 Pathak, D. (2024). 腿式机器人的极限跑酷. 在国际会议中. Fu, Z., Kumar, A., Malik, J., 和 Pathak, D. (2021). 最小化能量消耗导致腿式机器人步态的出现. 在 机器学习会议中. PMLR. Haarnoja, T., Zhou, A., Abbeel, P., 和 Levine, S. (2018). Soft actor-critic: 基于随机行动者的离策略最大熵深度强化学习. 在国际机器学习会议中, 第1861–1870页. PMLR. Hansen, N. (2016). CMA进化策略：教程. *arXiv*预印本 *arXiv:1604.00772*. He, T., Zhang, C., Xiao, W., He, G., Liu, C., 和 Shi, G. (2024). 敏捷但安全：学习无碰撞的高速腿式运动. 在机器人学：科学与系统中. Hoyt, D. F. 和 Taylor, C. R. (1981). 马的步态与运动能量学. 自然, 292(5820):239–240. Huang, X., Chi, Y., Wang, R., Li, Z., Peng, X. B., Shao, S., Nikolic, B., 和 Sreenath, K. (2024). Diffuse-loco: 基于离线数据集的扩散模型实时腿式运动控制. 在 机器学习会议中. Humphreys, J., Li, J., Wan, Y., Gao, H., 和 Zhou, C. (2023). 四足运动的仿生步态转换. *IEEE*机器人与自动化快报. Hwangbo, J., Lee, J., Dosovitskiy, A., Bellicoso, D., Tsounis, V., Koltun, V., 和 Hutter, M. (2019). 学习腿式机器人的敏捷和动态运动技能. 科学机器人学, 4(26):eaau5872. Jacobs, R. A., Jordan, M. I., Nowlan, S. J., 和 Hinton, G. E. (1991). 局部专家自适应混合.神经计算, 3(1):79–87. Ji, G., Mun, J., Kim, H., 和 Hwangbo, J. (2022). 控制策略与状态估计器的并发训练以实现动态且稳健的腿式运动. *IEEE*机器人与自动化快报, 7(2):4630–4637. Kim, M. J., Pertsch, K., Karamcheti, S., Xiao, T., Balakrishna, A., Nair, S., Rafailov, R., Foster, E., Lam, G., Sanketi, P., 等. (2024). OpenVLA：一个开源的视觉-语言-动作模型. *arXiv*预印本 *arXiv:2406.09246*. Liang, B., Sun, L., Zhu, X., Zhang, B., Xiong, Z., Li, C., Sreenath, K., 和 Tomizuka, M. (2024). 自适应能量正则化用于自主步态转换和节能四足运动. *arXiv*预印本 *arXiv:2403.20001*.

Margolis, G. B. and Agrawal, P. (2023). Walk these ways: Tuning robot control for generalization with multiplicity of behavior. In *Conference on Robot Learning*, pages 22–31. PMLR.

Mitchell, A. L., Merkt, W., Papatheodorou, A., Havoutis, I., and Posner, I. (2024). Gaitor: Learning a unified representation across gaits for real-world quadruped locomotion. In *Conference on Robot Learning*. PMLR.

O’Mahoney, R., Mitchell, A. L., Yu, W., Posner, I., and Havoutis, I. (2024). Offline adaptation of quadruped locomotion using diffusion models. *arXiv preprint arXiv:2411.08832*.

Owaki, D. and Ishiguro, A. (2017). A quadruped robot exhibiting spontaneous gait transitions from walking to trotting to galloping. *Scientific reports*, 7(1):1–10.

Peng, X. B., Chang, M., Zhang, G., Abbeel, P., and Levine, S. (2019). Mcp: Learning composable hierarchical control with multiplicative compositional policies. *Advances in Neural Information Processing Systems*, 32.

Reske, A., Carius, J., Ma, Y., Farshidian, F., and Hutter, M. (2021). Imitation learning from mpc for quadrupedal multi-gait control. In *2021 IEEE International Conference on Robotics and Automation (ICRA)*, pages 5014–5020. IEEE.

Shafiee, M., Bellegarda, G., and Ijspeert, A. (2024). Viability leads to the emergence of gait transitions in learning agile quadrupedal locomotion on challenging terrains. *Nature Communications*, 15(1):3073.

Shao, Y., Jin, Y., Liu, X., He, W., Wang, H., and Yang, W. (2021). Learning free gait transition for quadruped robots via phase-guided controller. *IEEE Robotics and Automation Letters*.

Tang, Y., Yu, W., Tan, J., Zen, H., Faust, A., and Harada, T. (2023). Saytap: Language to quadrupedal locomotion. *arXiv preprint arXiv:2306.07580*.

Yang, C., Yuan, K., Zhu, Q., Yu, W., and Li, Z. (2020). Multi-expert learning of adaptive legged locomotion. *Science Robotics*, 5(49):eabb2174.

Yang, Y., Zhang, T., Coumans, E., Tan, J., and Boots, B. (2022). Fast and efficient locomotion via learned gait transitions. In *Conference on Robot Learning*, pages 773–783. PMLR.

Yu, W., Yang, C., McGreavy, C., Triantafyllidis, E., Bellegarda, G., Shafiee, M., Ijspeert, A. J., and Li, Z. (2023). Identifying important sensory feedback for learning locomotion skills. *Nature Machine Intelligence*, 5(8):919–932.

Zhuang, Z., Fu, Z., Wang, J., Atkeson, C., Schwertfeger, S., Finn, C., and Zhao, H. (2023). Robot parkour learning. In *Conference on Robot Learning*, pages 73–92. PMLR.

Margolis, G. B. 和 Agrawal, P. (2023)。Walk these ways: 通过行为多样性调优机器人控制以实现泛化。见机器学习会议，第22–31页。PMLR。Mitchell, A. L., Merkt, W., Papatheodorou, A., Havoutis, I., 和 Posner, I. (2024)。Gaitor: 学习跨步态统一表示以实现真实世界四足运动。见机器学习会议。PMLR。O’ Mahoney, R., Mitchell, A. L., Yu, W., Posner, I., 和 Havoutis, I. (2024)。使用扩散模型进行四足运动的离线适应。*arXiv*预印本 *arXiv:2411.08832*。Owaki, D. 和 Ishiguro, A. (2017)。一种从行走过渡到小跑再到疾驰表现出自发步态转换的四足机器人。科学报告，7(1): 1–10。Peng, X. B., Chang, M., Zhang, G., Abbeel, P., 和 Levine, S. (2019)。Mcp: 使用乘法组合策略学习可组合分层控制。神经信息处理系统进展，32。Reske, A., Carius, J., Ma, Y., Farshidian, F., 和 Hutter, M. (2021)。基于MPC的模仿学习用于四足多步态控制。见2021年*IEEE*国际机器人与自动化会议（*ICRA*），第5014–5020页。IEEE。Shafiee, M., Bellegarda, G., 和 Ijspeert, A. (2024)。可行性导致在具有挑战性的地形上学习敏捷四足运动时步态转换的出现。自然通讯，15(1): 3073。Shao, Y., Jin, Y., Liu, X., He, W., Wang, H., 和 Yang, W. (2021)。通过相位引导控制器学习四足机器人的自由步态转换。*IEEE*机器人与自动化快报。Tang, Y., Yu, W., Tan, J., Zen, H., Faust, A., 和 Harada, T. (2023)。Saytap: 语言到四足运动。*arXiv*预印本 *arXiv:2306.07580*。Yang, C., Yuan, K., Zhu, Q., Yu, W., 和 Li, Z. (2020)。自适应腿式运动的多专家学习。科学机器人学，5(49): eabb2174。Yang, Y., Zhang, T., Coumans, E., Tan, J., 和 Boots, B. (2022)。通过学习到的步态转换实现快速高效运动。见机器学习会议，第773–783页。P MLR。Yu, W., Yang, C., McGreavy, C., Triantafyllidis, E., Bellegarda, G., Shafiee, M., Ijspeert, A. J., 和 Li, Z. (2023)。识别学习运动技能的重要感觉反馈。自然机器智能，5(8): 919–932。Zhuang, Z., Fu, Z., Wang, J., Atkeson, C., Schwertfeger, S., Finn, C., 和 Zhao, H. (2023)。机器人跑酷学习。见机器学习会议，第73–92页。PMLR。